

Computer Networks – Capturing and Analysing Packets Using Wireshark

Aim: To capture and analyse network packets using the Wireshark tool.

Background

Wireshark is a network packet analysis tool used to capture and study network traffic. It displays detailed information about different network protocols, such as TCP, UDP, ARP, DNS, HTTP, ICMP, and DHCP.

Requirements

- Computer system
- Wireshark software installed
- Active network connection
- Ethernet / LAN interface

Procedure

1. Open the **Wireshark** application.
2. Select the active network interface (e.g., **Local Area Connection / Ethernet**).
3. Navigate to **Capture → Options**.
4. Set the capture stop condition limit to **100 packets**.
5. Start the live packet capture.
6. Apply protocol display filters in the filter bar to isolate specific traffic:

Display Filter	Target Traffic
<code>tcp</code>	TCP packets
<code>udp</code>	UDP packets
<code>arp</code>	ARP packets
<code>dns</code>	DNS packets
<code>http</code>	HTTP packets

Display Filter	Target Traffic
icmp	ICMP packets
dhcp	DHCP packets

7. Observe and inspect packet header details and payloads in the packet list and packet details panes.
8. Save the captured trace file (.pcapng) for documentation and future reference.

Result: Thus, network packets were successfully captured and analysed using Wireshark.